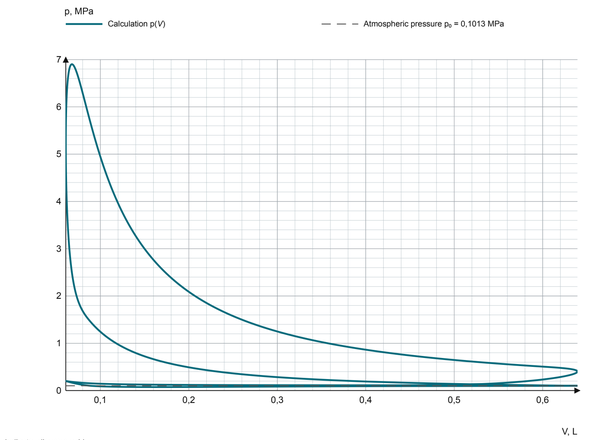
Vehicle powertrain design calculation

ICE Design 0.8.6 · 110 kW specified / 110 kW calculated. *Maximum torque is determined from the calculated speed grid.

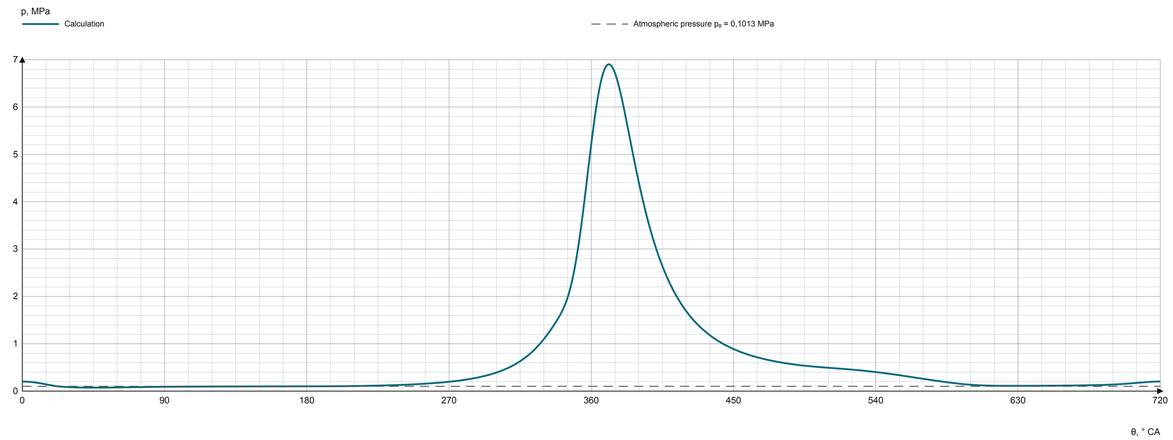
Powertrain architecture **Internal combustion engine**	I-2 · Ignition **Spark ignition (petrol engine)**	I-6 · Required power **110 kW**	I-7 · Shaft speed **5 500 min⁻¹**	I-8 · Compression ratio **10,5**	I-9 · Arrangement **In-line**	I-10 · Chamber **Compact undivided**
I-11 · Mixture preparation **Port fuel injection (PFI)**	I-12 · Intake (absolute) **101,3 kPa**	I-13 · Compressor **No compressor**	I-14 · Cooling **Liquid**	Cylinders / cycle **4 / 4 strokes**	D × S / displacement **91,86 × 87,27 mm / 2,314 L**	Maximum brake torque* **215,5 N·m at 1 513 min⁻¹**

Indicator diagram



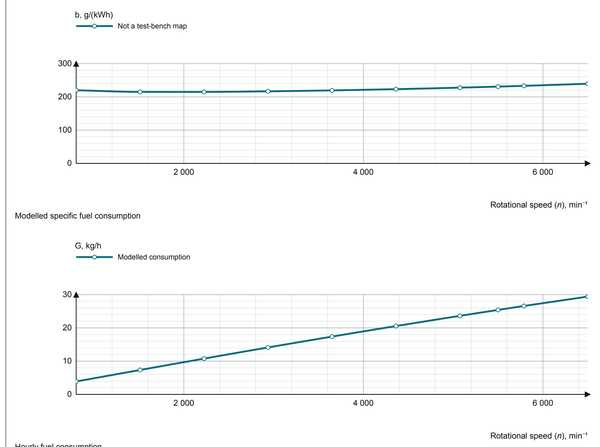
Indicator diagram p–V

Crank-angle indicator diagram p–θ 0–720° CA



Crank-angle indicator diagram p–θ, 0–720° CA

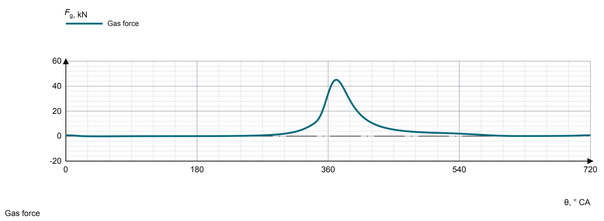
Fuel economy



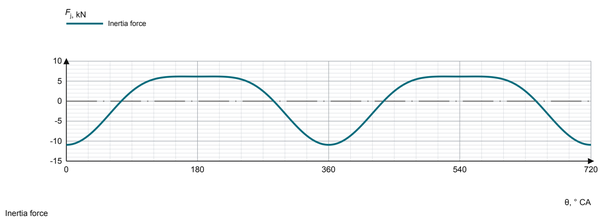
Modelled specific fuel consumption

Hourly fuel consumption

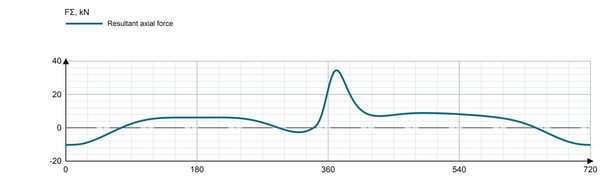
Axial forces: separately and combined



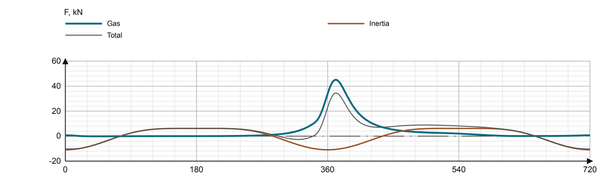
Gas force



Inertia force

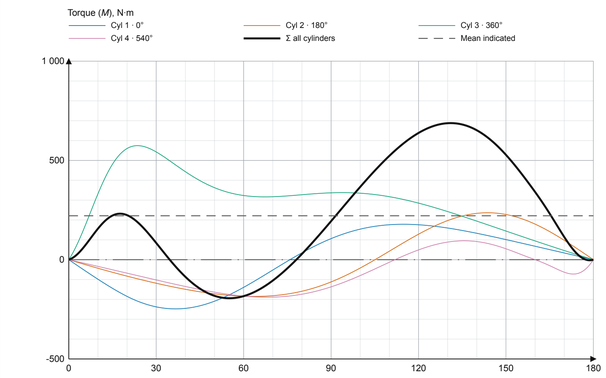


Resultant axial force

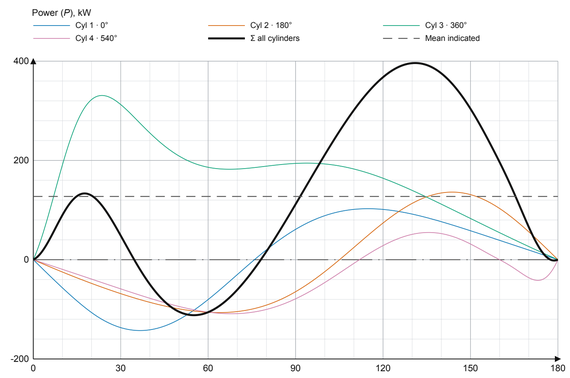


Axial forces of one cylinder

Torque and power over one firing interval



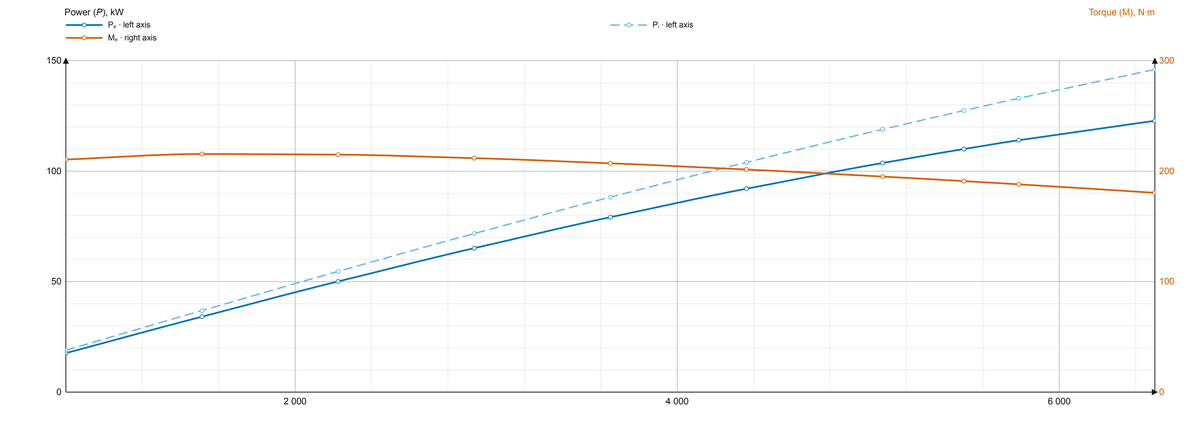
Cylinder contributions and total torque over one firing interval



Cylinder contributions and total instantaneous power over one firing interval

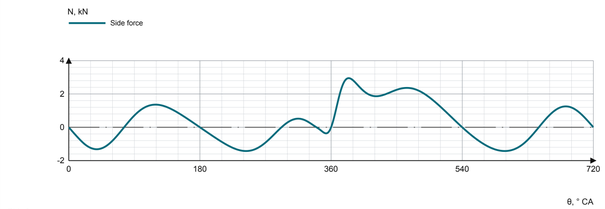
Δθ = 720° / 4 = 180°. Displays 0–Δθ; individual curves are fragments of the complete 720° cycle.
The sum repeats every Δθ for identical cylinders with even phase spacing. Instantaneous curves precede mechanical losses.
Mean M_i = 221.2 N·m; M_e = 191 N·m.

Complete-engine speed characteristic

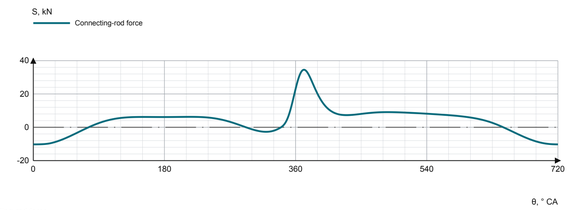


Full-load speed characteristic: power and torque

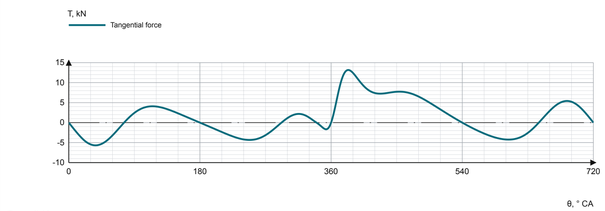
Separate connecting-rod and crank forces



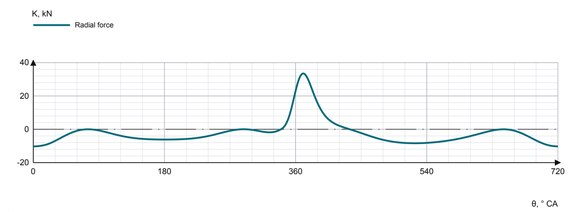
Side force



Connecting-rod force



Tangential force



Radial force

A1 841 × 594 mm · One sheet · Characteristics of the current variant.
Indicated cycle: 720°; total-curve period: 720°/i. M_i is mean indicated torque; M_e is mean brake torque. Model conditions and results, not test-bench measurements.